9 (a)(i) For the moon to orbit the Earth, $\frac{GMm}{r^2} = mr\left(\frac{2\pi}{T}\right)^2 \Rightarrow r^3 = \frac{GM}{4\pi^2}T^2$

$$\Rightarrow r = \left(\frac{GM}{4\pi^2}T^2\right)^{\frac{1}{3}} = \left(\frac{6.67\times10^{-11}\left(5.98\times10^{24}\right)}{4\pi^2}\left(27.4\times24\times3600\right)^2\right)^{\frac{1}{3}}$$

$$\therefore r = 3.84\times10^{8}\text{ m}$$

(ii) The total energy in elliptical orbits remains constant. By Conservation of Energy,

Maximum increase in KE of the moon = Maximum decrease in GPE of the moon,

$$\Delta KE = -\frac{GMm}{r_i} - \left(-\frac{GMm}{r_f}\right)$$

$$\Rightarrow \Delta KE = GMm\left(\frac{1}{r_f} - \frac{1}{r_i}\right)$$

$$= 6.67\times10^{-11}(5.98\times10^{24})(7.36\times10^{22})\left(\frac{1}{3.84\times10^{8}} - \frac{1}{4.07\times10^{8}}\right)$$

$$\therefore \Delta KE = 4.32\times10^{27}\text{ J}$$

(b)(i) For the satellite to orbit the Earth with radius R, the <u>gravitational force on the satellite provides the centripetal force</u>.

$$\frac{GM_E m}{R^2} = \frac{mv^2}{R}$$

$$\therefore KE = \frac{1}{2}mv^2 = \frac{GM_E m}{2R}$$

(ii) Total energy E of the satellite,

$$E = KE + PE = \frac{GM_E m}{2R} + \left(-\frac{GM_E m}{R}\right) = -\frac{GM_E m}{2R}$$

(iii) By Conservation of Energy,

KE at launch site + GPE = Total Energy at geosynchronous orbit

$$\Rightarrow \frac{1}{2}mv^2 + \left(-\frac{GM_E m}{R_E}\right) = -\frac{GM_E m}{2R}$$

$$\Rightarrow \frac{1}{2}(120)v^2 + \left(-\frac{6.67\times10^{-11}(5.98\times10^{24})(120)}{6.37\times10^{6}}\right) = -\frac{6.67\times10^{-11}(5.98\times10^{24})(120)}{2(3.6\times10^{7} + 6.37\times10^{6})}$$

$$\therefore v = 1.08\times10^{4}\text{ m s}^{-1}$$

(c)(i) From the graph, gravitation potential at 100 km, $\phi_{100km} = -61.52$ MJkg^{-1}

From the graph, gravitation potential at 80 km, $\phi_{80km} = -61.72$ MJkg^{-1}

$$\text{Increase in GPE } = m(\phi_{100km} - \phi_{80km})$$
$$= 4200(-61.52 - (-61.72)) \times 10^6$$
$$= 8.40 \times 10^8 \text{ J}$$

From (b)(ii), total energy $E = \dfrac{GPE}{2}$

Work done on the shuttle = Increase in total energy = (Increase in GPE)/2

= 4.20×10^8 J

(ii) Gravitational Field Strength, $g = -\dfrac{d\phi}{dr}$

From the graph, the gradient, $\dfrac{d\phi}{dr}$ is constant.

Therefore, gravitational field strength is constant.

(iii) From the graph, gradient, $\dfrac{d\phi}{dr} = \dfrac{(-61.52 - (-62.50)) \times 10^6}{(100-0) \times 10^3} = 9.8\text{Nkg}^{-1}$

Therefore, gravitational field strength g at 100 km is 9.8 N kg^{-1}.

(iv) For the space shuttle to orbit the Earth,

$$\frac{GMm}{r^2} = mr\left(\frac{2\pi}{T}\right)^2 \Rightarrow g = \frac{GM}{r^2} = r\left(\frac{2\pi}{T}\right)^2$$

$$\therefore T = 2\pi\sqrt{\frac{r}{g}} = 2\pi\sqrt{\frac{6.37 \times 10^6 + 100 \times 10^3}{9.8}} = 5.1 \times 10^3 \text{ s}$$

10(a)(i) e.m.f. is the amount of other forms of energy converted to electrical energy per unit